

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 8-9

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- Breusch-Pagan: Reject H_0 if $\frac{SSReg_{aux}}{2} > \chi^2_{\alpha,p}$.
- Standardized residual: $\frac{e_i}{s\sqrt{1-h_{ii}}}$
- Leverage: $h_{ii} = \mathbf{x}'_i(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$, $h_{avg} = \frac{p+1}{n}$
- SLR: $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2}$
- Cook's Distance: $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}} \right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- Variance Inflation: $VIF_j = \frac{1}{1-R_j^2}$, $j = 1, 2, \dots, p$

MLR Assumptions Recap

- Multiple linear regression model assumptions:
- 1. $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip} + \varepsilon_i$ is the model
- 2. $\{x_{i1}, x_{i2}, \dots, x_{ip}\}$ are **non-stochastic variables**.
- 3. $E[\varepsilon_i] = 0$
- 4. **Homoscedasticity**: $Var(\varepsilon_i) = \sigma^2$
- 5. y_i' s / ε_i' s are **independent**.
- 6. y_i' s / ε_i' s are **normally distributed**.
- 7. Predictors are **non-collinear**.

MLR Violations

- These mirror the regression assumptions.
- 1. **Mis-specified model:** _____
- 2. Residuals have _____
- 3. **Heteroscedasticity:** _____

MLR Violations 2

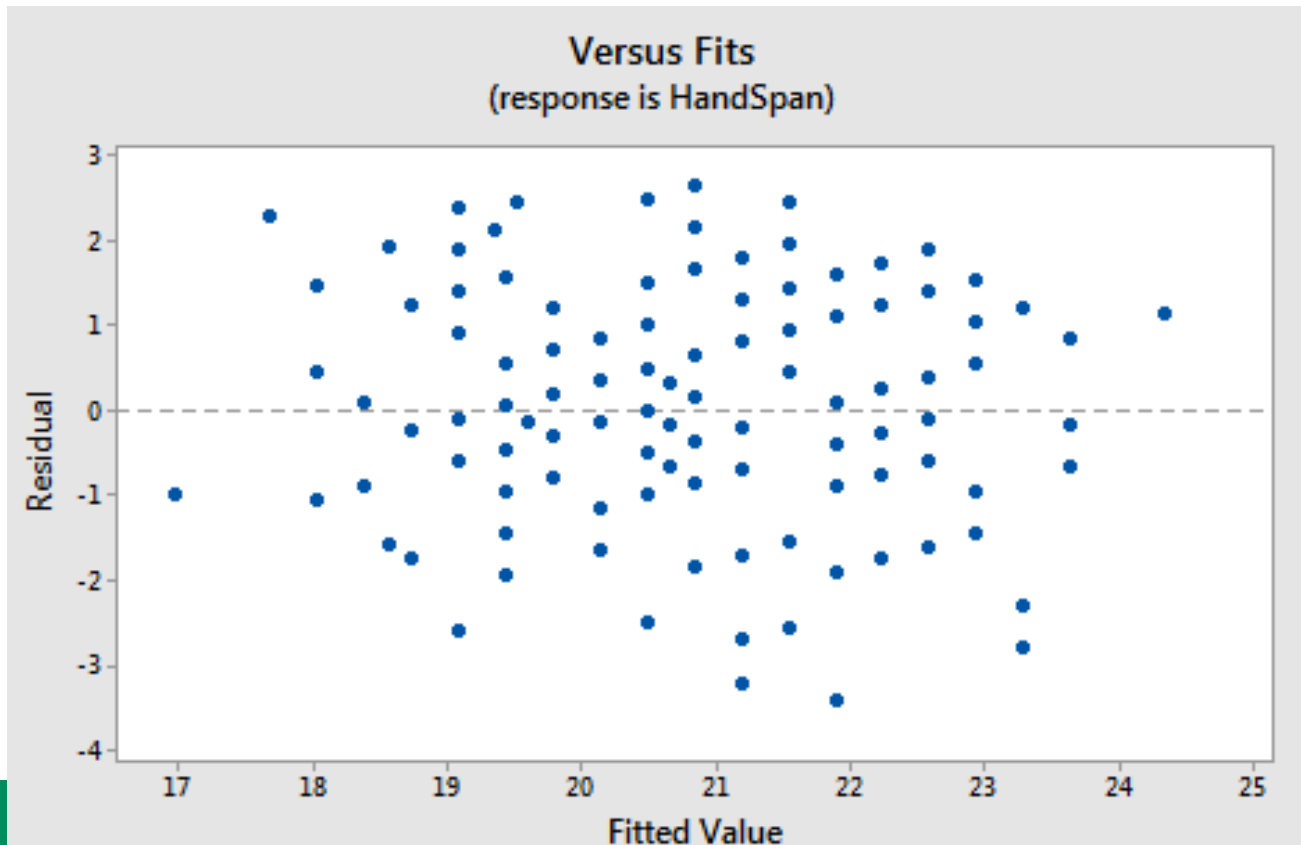
- 4. Residuals exhibit **serial correlation** : _____
- 5. **Non-normality** : _____

MLR Issues

- These are potential issues to investigate.
- **Outliers** –
- **High leverage points** –
- **Multi-collinearity** –
- **High dimensionality** –

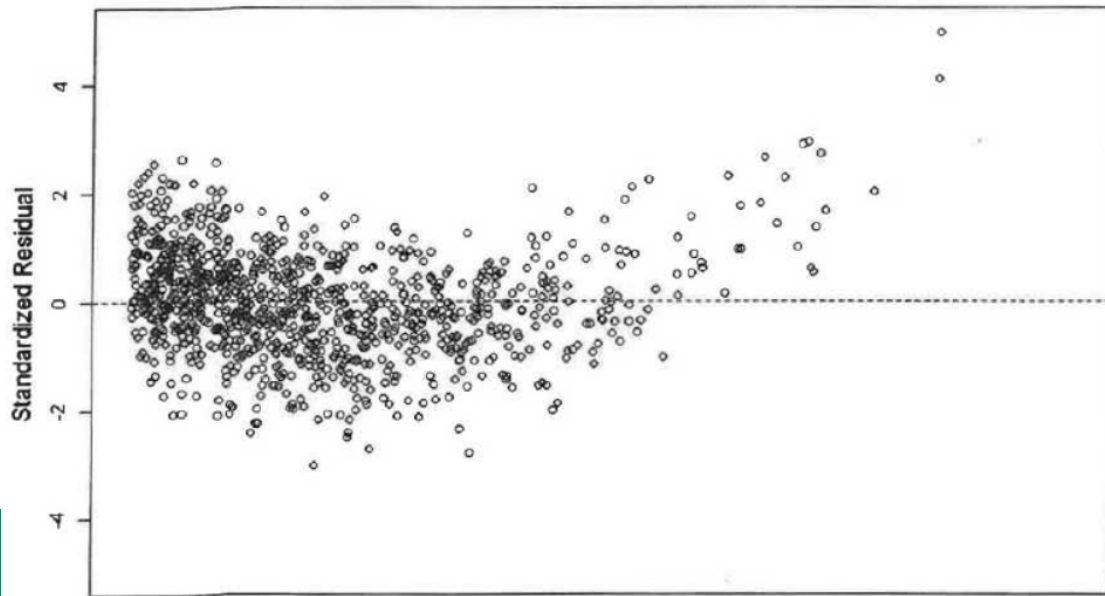
Good Residuals

- No discernable patterns, random scatter about 0 and average to 0, consistent spread



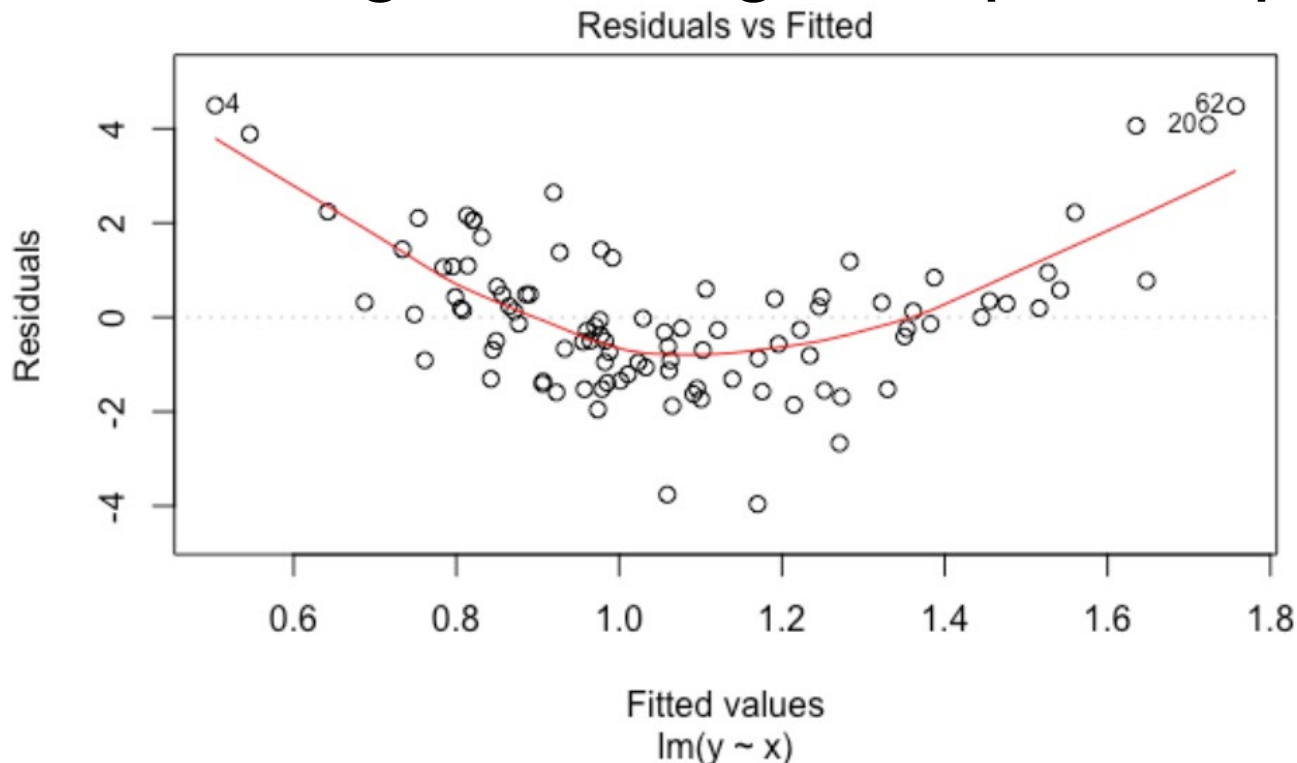
#1. Mis-Specified Model

- Residuals with a clear non-linear pattern may indicate the need to include additional terms.
- Underfitting Consequences:



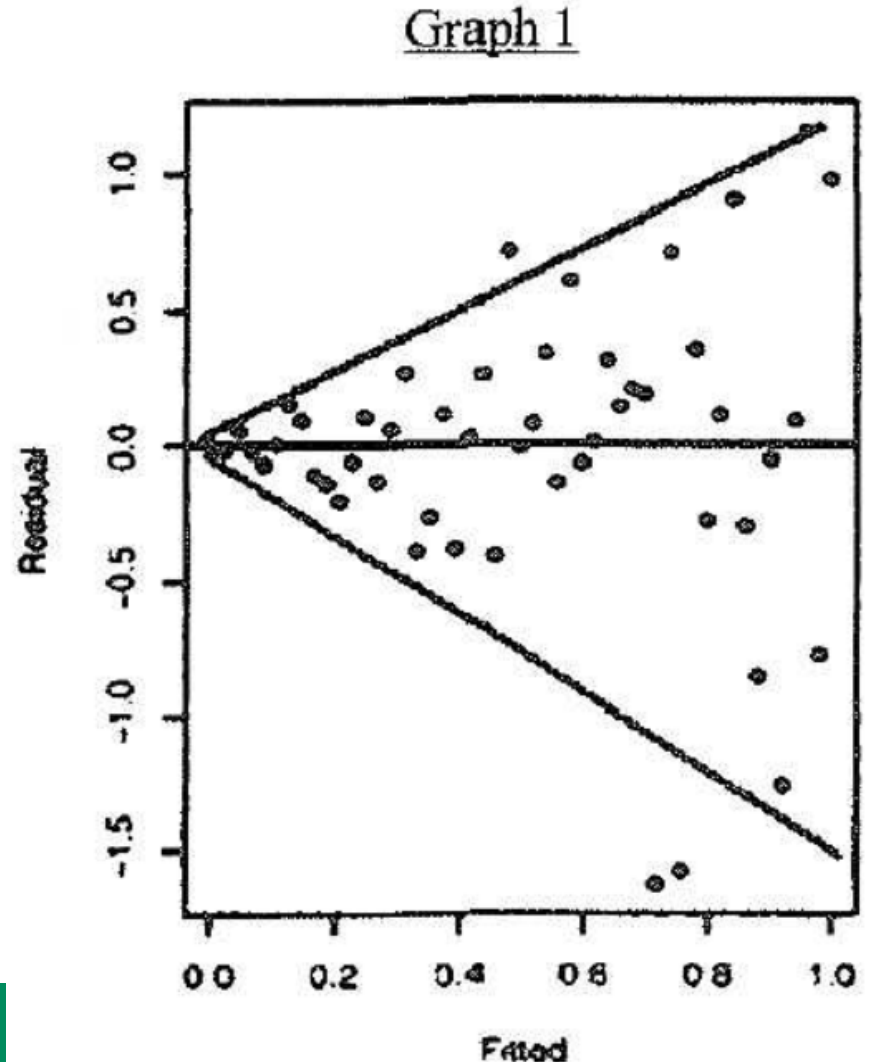
#2. Non-Zero Averages

- <https://thestatsgeek.com/2020/03/23/the-mean-of-residuals-in-linear-regression-is-always-zero/>
- Non-zero averages in ranges implies a pattern.



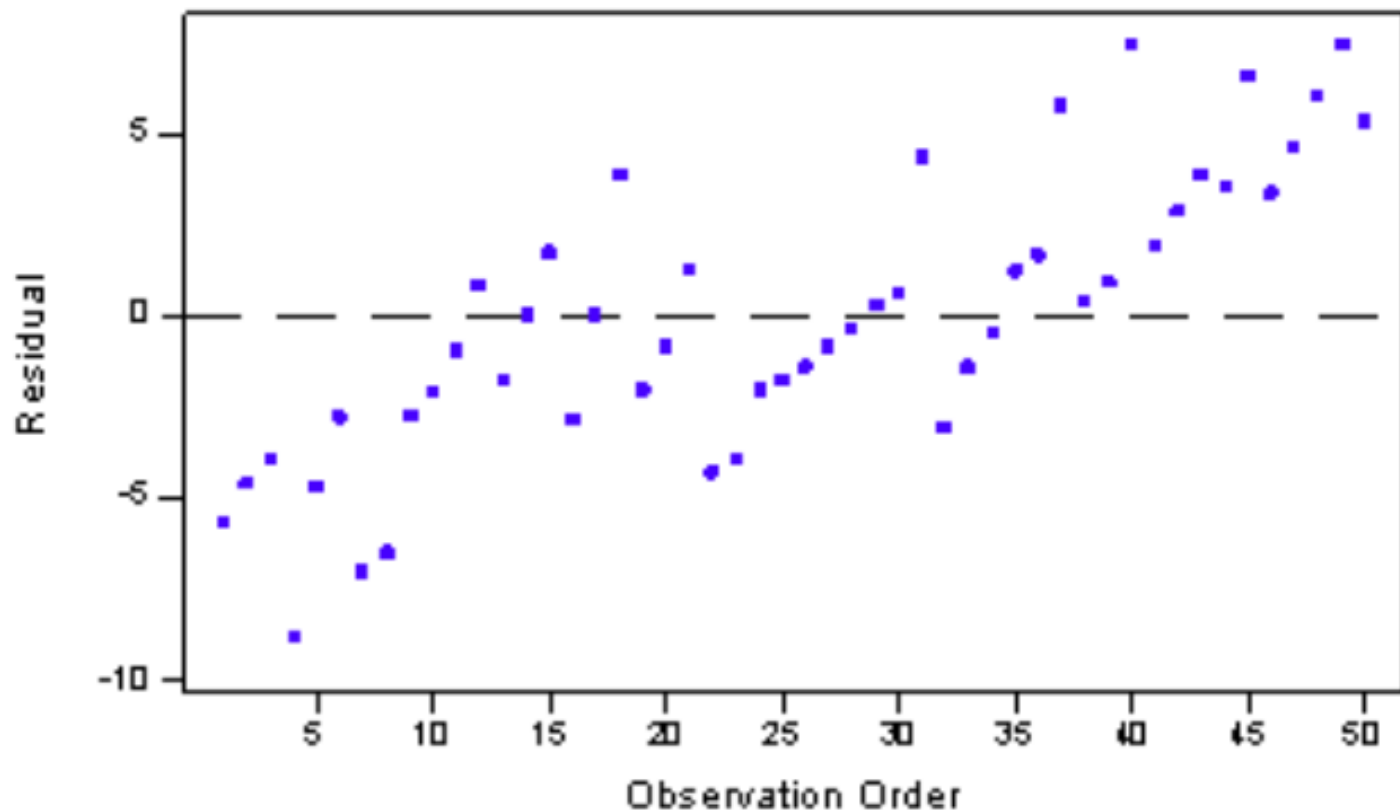
#3. Heteroscedasticity (CAS)

- Can ignore mild (still unbiased)
- Transforming Y may help
- Severe may require



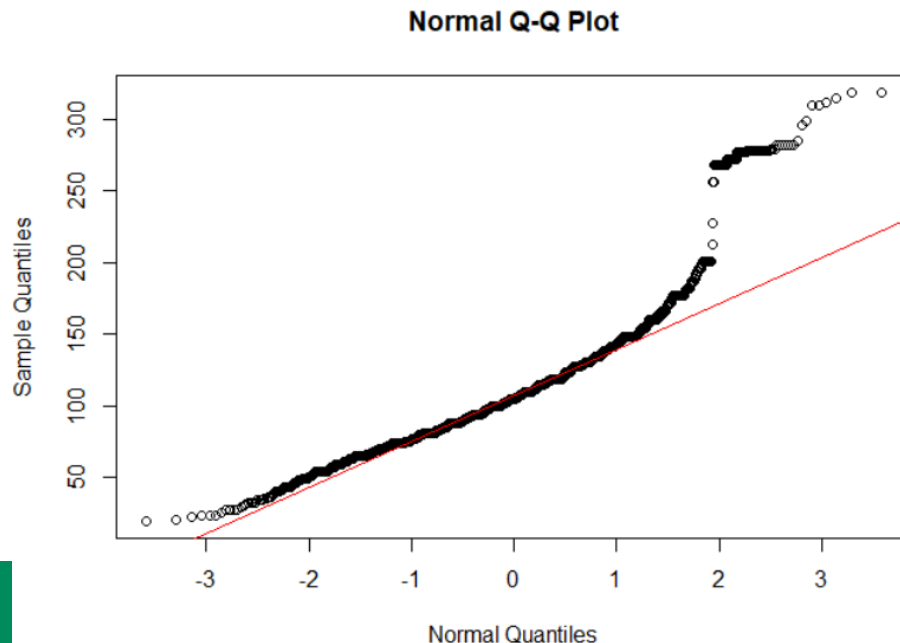
#4. Dependent Errors

- Serial correlation occurs when the residuals have a pattern _____



#5. Non-Normality

- Quantile-Quantile (**QQ Plots**): Measures data fit with reference model (typically normal)
- Follow the 45-degree reference line if normal
- Fixes: _____



Residual Plots

- A plot of e against $\hat{y}$ detects _____
- A plot of e against observation time detects _____
- A QQ-plot of e detects _____

Breusch-Pagan Test

- The Breusch-Pagan Test is hypothesis test method to test if heteroscedasticity exists in a linear regression model, assuming normally distributed error terms.
- 1. Calculate squared standardized residuals from the model _____
- 2. Run a regression with _____

Breusch-Pagan Test 2

- H_0 : Homoscedasticity in model
- H_a : Heteroscedasticity in model
- $LM =$
 - $SSReg_{aux}$: Regression sum of squares for the second model
- Reject H_0 if _____

1. B-P Illustration

- A linear regression model of Y (# of confusing trends) against X (# of books read) results in a mean-squared error of 25 and a F-statistic of 2 with 1 and 16 degrees of freedom.
- A linear regression model of the squared standardized residuals from the first model against X results in a $MSE = 5$ and $F_{1,16} = 0.5$.
- Use the Breusch-Pagan test at $\alpha = 5\%$ to test for heteroscedasticity.

1. B-P Illustration 2

- $LM = \frac{SSReg_{aux}}{2}$
- $F_{p,n-(p+1)} = \frac{\frac{SSReg}{p}}{MSE}$

Heteroscedasticity Fixes

- Transform the _____
- Use _____
- Adjust standard errors to _____
- FYI: $Var(\mathbf{b}) = \sum_{i=1}^n \mathbf{w}_i e_i^2 \mathbf{w}_i^T$ where $\mathbf{w}_i = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i$

Weighted Least Squares

- In weighted least squares, the response becomes $\sqrt{w_i}y$ and the predictors $\sqrt{w_i}x$.
- $Var(\varepsilon_i) = \frac{\sigma^2}{w_i}$
 - w_i : observation weight
- Goal: Minimize $\sum_{i=1}^n w_i (y_i - \hat{y}_i)^2$
- FYI: $b = (X'WX)^{-1}X'W\mathbf{y}$

Leverage

- **Influential points** have a strong effect on the overall model fit.
- High **leverage** points have an unusual _____

Leverage 2

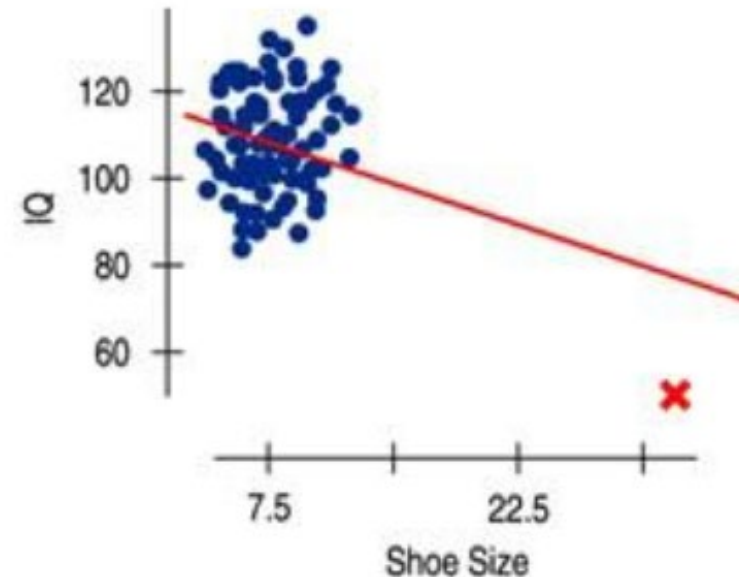
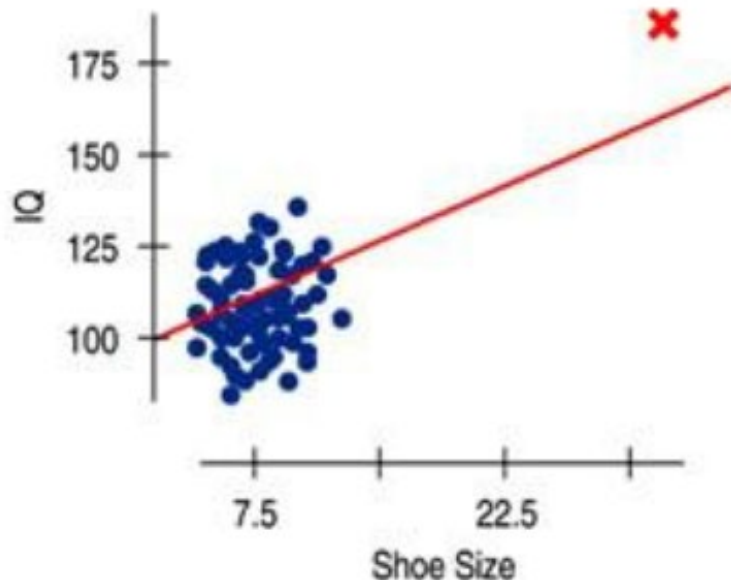
- $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H} \mathbf{y}$ where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$
 - $\mathbf{H}$: hat matrix (puts the hat on $\mathbf{y}$)
- $\widehat{Var}(e_i) = s^2(1 - h_{ii})$
- $Cov(e_i, e_j) = -s^2(Cov_{ij} \text{ term from } \mathbf{H})$

Leverage 3

- $\hat{y}_i = h_{i1}y_1 + h_{i2}y_2 + \cdots + h_{ii}y_i + \cdots h_{in}y_n$
 - h_{ij} : Row i , Column j entry from hat matrix $\mathbf{H}$
- Simple linear regression: $h_i =$
 - $\frac{1}{n} \leq h_{ii} \leq 1$ always
 - $h_{avg} = \frac{p+1}{n}$ with
 - p : # of explanatory predictors
 - n : # of data points
- High leverage point is

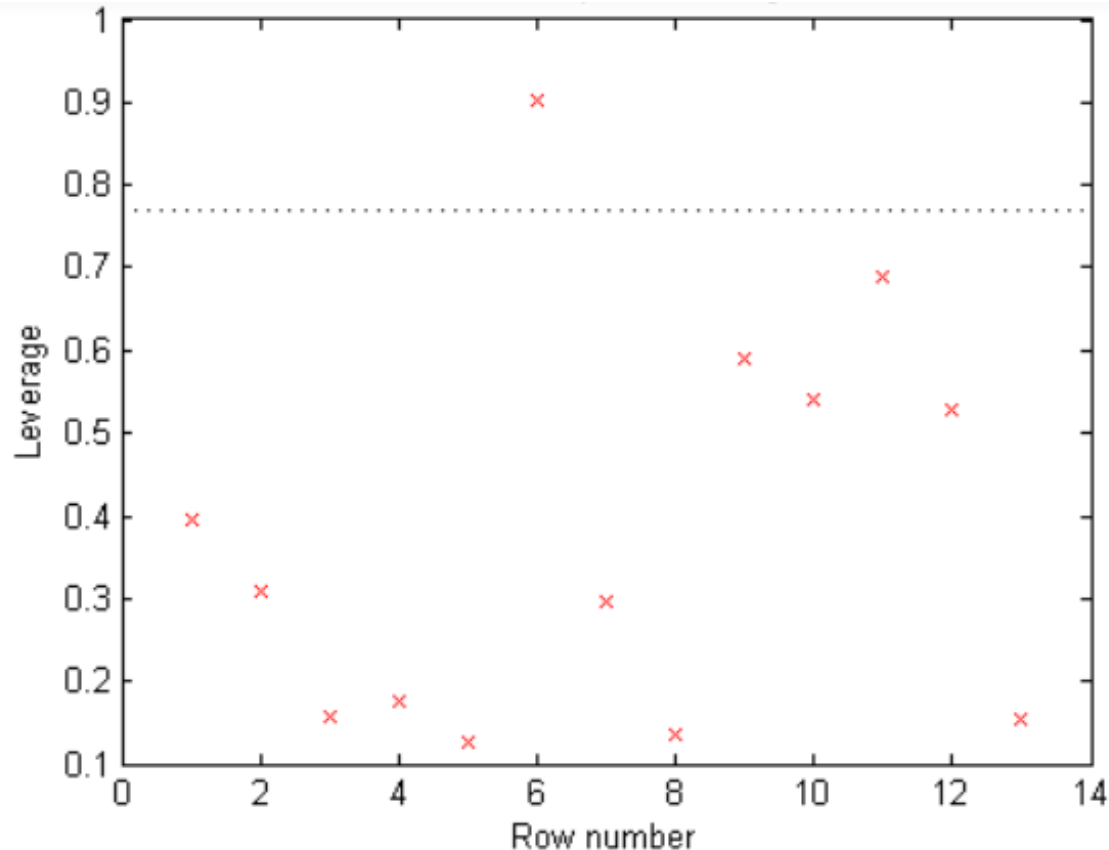
High Leverage Points

- <https://slideplayer.com/slide/3599913/>



Leverage Plot

- *Sample leverage plot*
- Residual, QQ plots do not detect high leverage points.



2. Leverage Illustration

- A basic linear regression model is fit to 100 observations. If $\sum_i x_i = 800$ and $\sum_i x_i^2 = 30,000$, calculate the leverage when $x = 5$.

2. Leverage Illustration 2

$$\bullet h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^n (x_j - \bar{x})^2} = \frac{1}{n} + \frac{\left(x_i - \frac{\sum_j x_j}{n}\right)^2}{\sum_j (x_j^2) - \frac{(\sum_j x_j)^2}{n}}$$

Leverage Handling

- Ways to handle high leverage points include:
- Include the observation & comment.
- Delete the observation if not representative, or present model _____
- Choose another explanatory variable to serve as a _____

Leverage Handling 2

- Transform the explanatory variable.
- Use **robust estimation** techniques that reduce effect of influential points.

Standardized Residuals

- Standardizing residuals makes them dimension-less.
- Standardized residual =
- $e_i = y_i - \hat{y}_i$
- $se(e_i) = s\sqrt{1 - h_{ii}}$
 - s^2 : Mean squared error (MSE)
 - h_{ii} : i^{th} diagonal element of hat matrix

Studentized Residuals

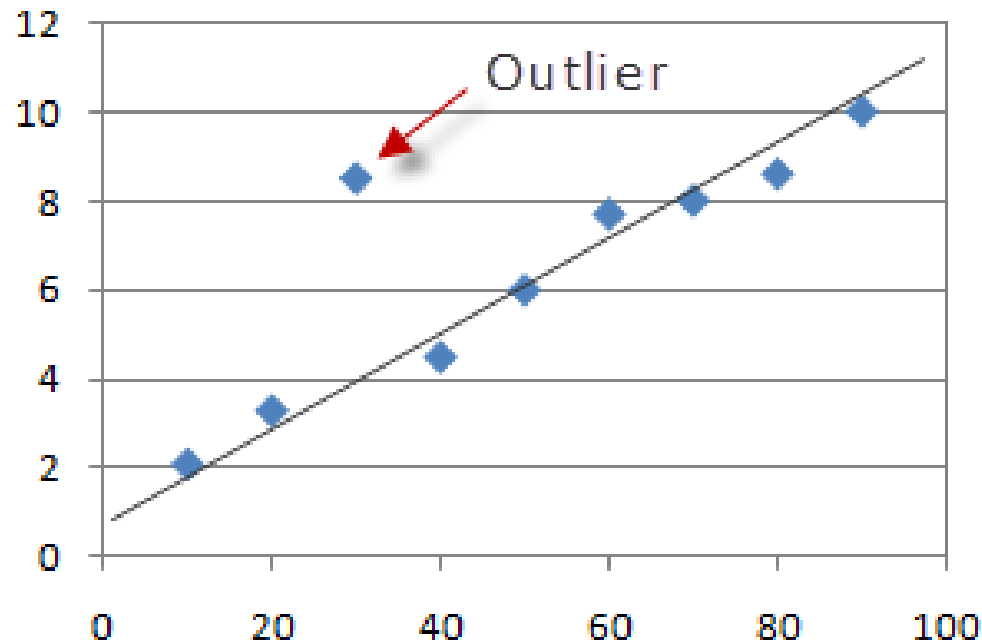
- **Studentized residuals** follow the same formula as the standardized residual, but the _____
- Studentized residuals are _____

Outliers

- Outliers are observations with a large residual, indicating it is not well fit by the model. They should be investigated.
- Typically, $|\text{standardized or studentized residuals}| > 2$ is considered an outlier.

Outlier Plot

- <https://www.mathopenref.com/outlier.html>



Handling Outliers

- Options for outliers:
- Include the observation but comment on it
- Delete the observation
- Create a binary variable to flag the outlier. If a cause drives the outlier, all points related to the cause can be flagged.

Concept Check #1

- Which statements are true?
- I. If residuals exhibit a predictable pattern from ordered observation to observation, transform the response to fix.
- II. The total leverage sums to the # of explanatory variables.
- III. Dependent MLR error terms lead to too high estimated standard errors.
- IV. If residuals are negative on the extremes and positive in the middle, adding a quadratic predictor may help.
- V. For simple linear regression with 10 observations, leverage above 0.6 is considered high.
- VI. High dimensional data makes R^2 , the fitted model, and parameter confidence intervals all unreliable.
- VII. A log or square root transformation can make the variance of the residuals less constant.

Example 1

- A basic linear regression model to predict # of daily tantrums by a child is fit to 4 observations.

- $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \begin{pmatrix} 0.6 & -0.2 & 0.1 & 0.3 \\ -0.2 & 0.3 & -0.1 & 0.5 \\ 0.1 & -0.1 & 0.4 & 0.01 \\ 0.3 & 0.5 & 0.01 & 0.7 \end{pmatrix}$

- $\mathbf{y} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 6 \end{pmatrix}$ and $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{pmatrix} 2.2 \\ 3.3 \\ 1.46 \\ 7.04 \end{pmatrix}$

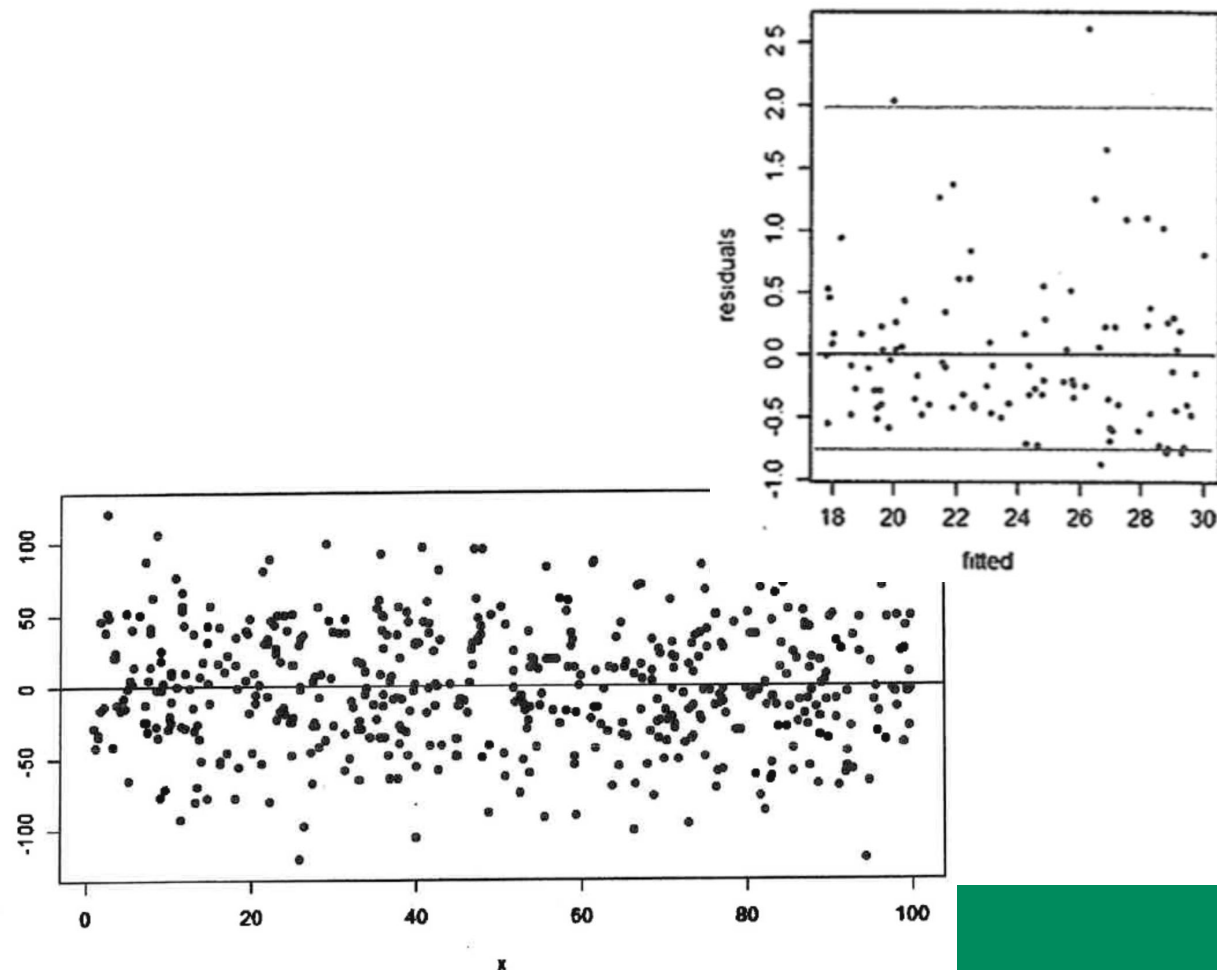
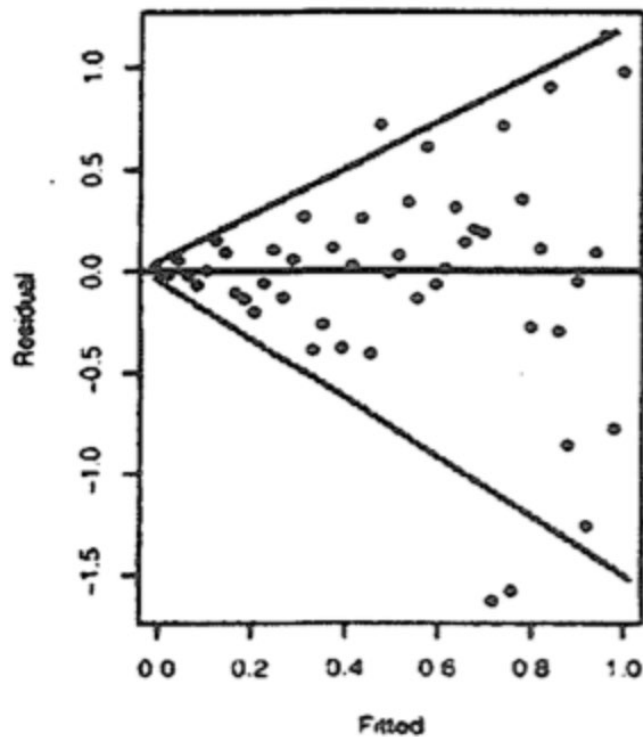
- $s^2 = 2$
- Calculate the standardized residual of the second observation. Is it an outlier?

Example 1 Sol

- Standardized residual = $\frac{e_i}{se(e_i)}$
- $e_i = y_i - \hat{y}_i$, $se(e_i) = s\sqrt{1 - h_{ii}}$, $MSE = s^2$
- $\hat{y}_i = h_{i1}y_1 + h_{i2}y_2 + \cdots + h_{ii}y_i + \cdots h_{in}y_n$

Concept Check #2

- For each graph, what MLR issues, if any, do they indicate?



Cook's Distance

- **Cook's distance** is a nice metric that quantifies the _____
- Average D_i is roughly $\frac{1}{n}$, with _____
- True comparison is to a F -statistic with $p + 1, n - (p + 1)$ degrees of freedom

Cook's Distance 2

- $$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{(p+1)s^2} =$$

- $\hat{y}_{j(i)}$: prediction of j^{th} observation with i^{th} observation left out of model fit
- p : # of explanatory predictors

3. Cook's D Illustration

- You fit a multiple linear regression model using 2 explanatory variables to predict the # of grumpy neighbors you have.
- Calculate leverage of the first observation and Cook's distance for the second observation.

Obs. #	y_i	$\hat{y}_i$	h_{ii}	$\frac{e_i}{se(e_i)}$	D_i
1	10	12		-0.5	0.125
2	20	19	0.3		

3. Cook's D Illustration 2

- $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}} \right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- $e_i = y_i - \hat{y}_i, se(e_i) = s\sqrt{1-h_{ii}}$

Example 2 Data

- The model $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ is fit to the data:

- $(X^T X)^{-1} X^T \mathbf{y} = \begin{bmatrix} 5.02 \\ -0.22 \\ -0.59 \end{bmatrix}$

- $X(X^T X)^{-1} X^T \mathbf{y} = \begin{bmatrix} 2.55 \\ 1.26 \\ \text{Missing} \\ 3.195 \\ 4.32 \end{bmatrix}; \mathbf{y} = \begin{bmatrix} \text{Missing} \\ 2 \\ 3 \\ \text{Missing} \\ \text{Missing} \end{bmatrix}$

- $X(X^T X)^{-1} X^T = \begin{bmatrix} 0.35 & 0.23 & -0.03 & 0.42 & 0.03 \\ 0.23 & 0.87 & 0.16 & -0.1 & -0.16 \\ -0.03 & 0.16 & 0.55 & -0.13 & 0.45 \\ 0.42 & -0.1 & -0.13 & 0.68 & 0.13 \\ 0.03 & -0.16 & 0.45 & 0.13 & 0.55 \end{bmatrix}$

Example 2 Qs

- A. Identify all the matrices.
- B. How many are high leverage points according to the rule of thumb?
- C. If the sum of squared residuals is 4.52 and the Cook's distance for the first point is 0.2935, what is the first residual (if negative)?
- D. What is the second standardized residual?
- E. If $x_{1,3} = 0.75$, $x_{2,3} = 2$ for the third observation, what is the third Cook's distance?
- F. (Bonus) What are the residuals for the 4th and 5th observations if $e_4 > e_5$?

Example 2 Sol

- $D_i = \left(\frac{e_i}{s\sqrt{1-h_{ii}}} \right)^2 \frac{h_{ii}}{(p+1)(1-h_{ii})}$
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- High Leverage: $\frac{3(p+1)}{n}$
- $e_{std,i} = \frac{e_i}{s\sqrt{1-h_{ii}}}$
- $e_i = y_i - \hat{y}_i$

Collinearity

- **Collinearity**/multicollinearity occurs when an explanatory variable is, or almost is, a linear combination of other available explanatory variables.
- If perfect collinearity exists, if all those variables are used, there are an _____
- Near collinearity does not prevent model fitting, but leads to _____

Collinearity 2

- Collinearity relates to predictor-predictor relationships, not _____
- The F -test is not affected by multi-collinearity.

Collinearity Detection

- A correlation matrix of predictors or pairwise scatterplots can help detect collinearity (but can miss collinearity among 3+ predictors).
- Residual, leverage, Cook's Distance plots will not detect collinearity.

VIF

- The **variance inflation factor** (VIF) is a measure of collinearity of an explanatory variable. **Tolerance** is the inverse of VIF.
- $VIF_j =$
 - VIF_j : VIF for j^{th} explanatory variable
 - R_j^2 : Coefficient of determination when x_j is used as the response against all other x' s
- $VIF_j \geq 10$ indicates severe collinearity
- $5 < VIF_j < 10$ indicates mild collinearity

Special Variable Terms

- **Orthogonal** predictors are completely _____
- A **suppressor variable** is an explanatory variable _____

Concept Check #3

- Which statements are true?
- I. The leverage for each observation of the linear model is between $\frac{1}{n}$ and 1.
- II. If the regression of X_1 on X_2 yields $R^2 = 0$, the two variables are orthogonal.
- III. The VIF for an uncorrelated variable is 0.
- IV. Heteroscedasticity leads to an unreliable VIF.
- V. Under perfect multi-collinearity, regression coefficients cannot be uniquely estimated.
- VI. Dependent MLR error terms lead to multi-collinearity.
- VII. If two variables are non-significant using t -tests, but the F -tests suggests at least one is significant, the variables could be negatively correlated and the model may be experiencing collinearity.
- VIII. Collinearity always implies one variable is redundant in the presence of another variable.

4. VIF Illustration

- In a multiple linear regression model for predicting alien life on other planets using 2 explanatory variables $\{x_1, x_2\}$,
- $\sum (x_{i1} - \bar{x}_1)^2 = 3,125$
- $\sum (y - \bar{y})^2 = 5,900$
- For a regression of $y \sim x_1 + x_2$, the error sum of squares is 500.
- For a regression of $x_1 \sim x_2$, the error sum of squares is 625.
- Calculate the variance inflation factor for x_1, x_2 .

4. VIF Illustration 2

- $VIF_j = \frac{1}{1-R_j^2}$
- $R^2 = \frac{SSR}{SST}$ and $SST = SSE + SSR$
- $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

Collinearity Handling

- Options to deal with high collinearity:
- Standardize _____
- Accept and discuss its implications.
- Transform into _____

Collinearity Handling 2

- Replace a collinear variable with a _____
- Remove one or more variables.
- Combine _____

Collinearity & Leverage

- Both collinearity and leverage are used for model criticism and based only on the _____
- Both naturally exist and do not inherently make a deficient model, but rather are useful for _____
- Collinearity can be masked or conversely created by high leverage points.

Issues Summary

Cause	Effect
Mis-specified model	Unreliable $b's$, MSE, $se's$, inference metrics
Heteroscedasticity	Unreliable $b's$, MSE
Dependent errors	Unreliable $b's$, $se's$ too low
Non-normality	Unreliable inference metrics
Outliers	Unreliable MSE
High leverage points	Unreliable $b's$
Multi-collinearity	Unreliable $b's$, $se's$ too high
High dimensionality	Unreliable $b's$, MSE

Concept Check #4

- Which of the following actions can be taken to address multi-collinearity?
- I. Do nothing
- II. Use orthogonal predictors
- III. Transform the response variable
- IV. Remove predictors
- V. Standardize predictors
- VI. Combine predictors

Example 3

- Given the following info, determine which predictors might be good to drop from the model. Predictors should not be removed from the model solely based on the variance inflation factor.

Predictor	VIF	95% Confidence Interval
x_1	3	$(-3.1, 0.5)$
x_2	7	$(-7.2, 5.9)$
x_3	8	$(-4.4, 0.1)$

Example 3 Sol

Example 4 (CAS)

- Two different data sets were used to construct the four regression models below.
- A. How collinear is dataset A?
- B. What range of SSE would make dataset B mildly but not severely collinear?

Data Set	Model	Dependent Variables	Independent Variables	SST	SSE
A	1	Y_A	X_{A1}, X_{A2}	36,000	2,800
A	2	X_{A1}	X_{A2}	9,000	600
B	3	Y_B	X_{B1}, X_{B2}	30,000	?
B	4	X_{B1}	X_{B2}	80,000	??

Example 4 Sol

- $VIF_j = \frac{1}{1-R_j^2}$
- $R^2 = \frac{SSR}{SST}$ and $SST = SSE + SSR$

HW Note

- Some Model Selection concepts are mixed into part of Model Assumption Qs.
- Validation/hold-out set validation fits 1 model (computationally cheapest).
- k-fold cross-validation fits k models.
- LOOCV is k-fold with $k = n$ models.

Summary Check

- Linear Model Violations (Mis-specified Model, Non-Zero Residual Averages, Heteroscedasticity, Dependent Errors, Non-Normality) and any potential fixes
- Linear Model Issues (Outliers, High Leverage, Collinearity, High Dimensions) and any potential fixes
- Formulas and properties for Breusch-Pagan, leverage, standardized residuals, Cook's distance, VIF